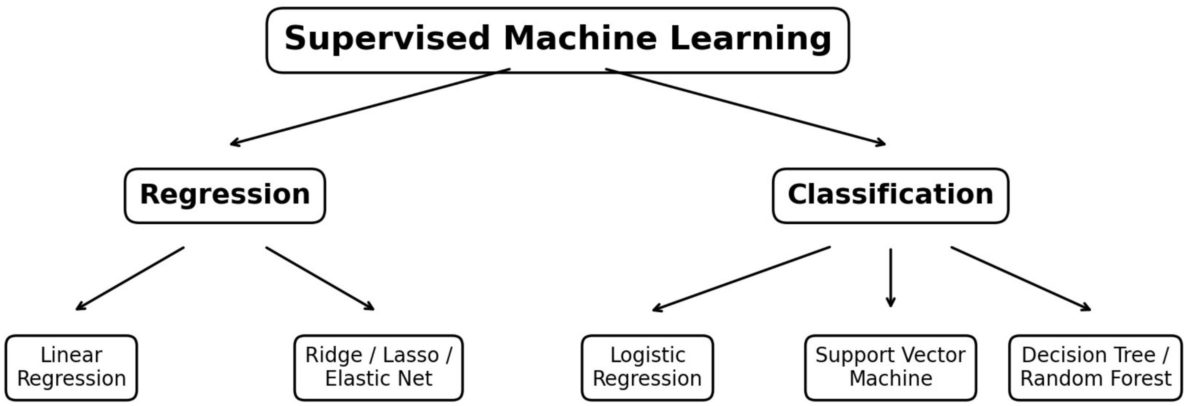
Machine Learning Study Notes

Regression, classification, validation, support vector machines, decision trees and random forests



Purpose — A corrected and organized fair copy of the handwritten notes. The formulas use standard notation and the explanations focus on exam-ready understanding.

Core idea: choose a model, define a loss function, train it by minimizing loss, validate it on unseen data, and control overfitting.

1. Linear Regression

Linear regression predicts a continuous numerical target by fitting a straight line or a linear hyperplane to the data.

Model

$$\textbf{Simple model: } \hat{y} = wx + b$$

$$\textbf{Multiple features: } \hat{y} = w^{T}x + b = \theta_0 + \theta_1 x_1 + \cdots + \theta_n x_n$$

- $\hat{y}$ is the predicted value; y is the observed value.
- w or θ are model coefficients; b or θ_0 is the intercept.
- The residual (error) for one observation is $e_i = \hat{y}_i - y_i$.

Mean Squared Error (MSE) cost

$$J(w,b) = (1 / 2m) \sum_{i=1}^{m} (\hat{y}_i - y_i)^2$$

- Squaring prevents positive and negative residuals from cancelling.
- Large errors receive a stronger penalty than small errors.
- The factor 1/2 is optional; it simplifies derivatives during optimization.

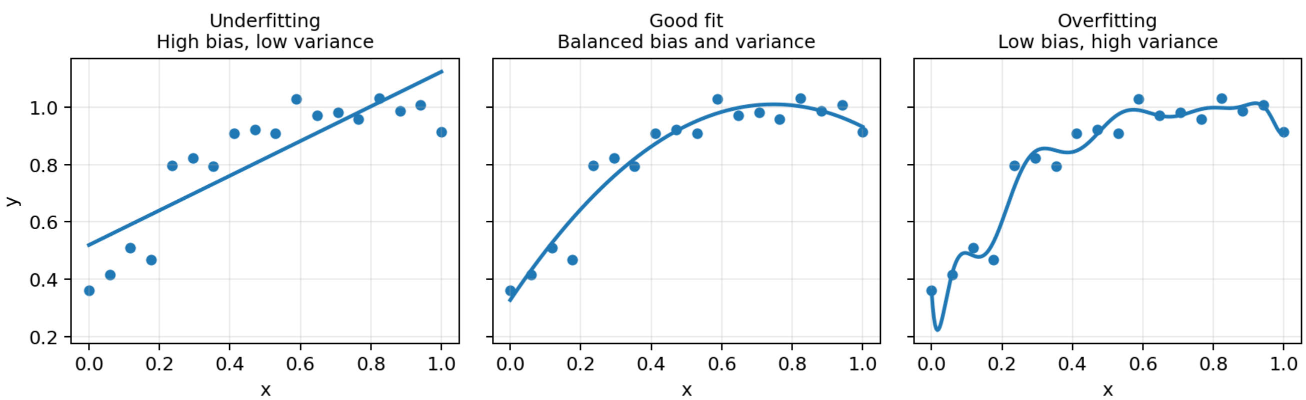
Gradient descent

$$w \leftarrow w - \alpha\,(\partial J/\partial w) \qquad \textbf{and} \qquad b \leftarrow b - \alpha\,(\partial J/\partial b)$$

Term	Meaning
α	Learning rate: step size used during each update.
Gradient	Direction of the steepest increase in loss.
Negative gradient	Direction used to reduce the loss.
Convergence	The loss stops decreasing meaningfully and parameters stabilize.

Learning-rate warning — A very large α can overshoot or diverge; a very small α makes training slow.

2. Model Fit: Underfitting and Overfitting



Situation	Typical pattern	Bias	Variance	Generalization
Underfitting	Model is too simple to capture the relationship.	High	Low	Poor on training and test data

Situation	Typical pattern	Bias	Variance	Generalization
Good fit	Model captures the main signal without memorizing noise.	Balanced	Balanced	Good on unseen data
Overfitting	Model learns training noise and unnecessary detail.	Low	High	Training performance high; test performance poor

- Bias describes systematic error caused by overly restrictive assumptions.
- Variance describes sensitivity to changes in the training sample.
- The aim is not simply maximum training accuracy; it is reliable performance on unseen data.

Regularization

Regularization adds a penalty to the original loss to discourage excessively large coefficients and reduce overfitting.

$$\textbf{Regularized objective = Original loss + Penalty}$$

Method	Penalty	Main effect	Feature selection?
Ridge (L2)	$\lambda \sum_{j=1}^{n} \theta_j^2$	Shrinks coefficients smoothly; useful with correlated features.	Usually no
Lasso (L1)	$\lambda \sum_{j=1}^{n} \lvert\theta_j\rvert$	Can force some coefficients exactly to zero.	Yes
Elastic Net	$\lambda_1 \Sigma\lvert\theta_j\rvert + \lambda_2 \Sigma\theta_j^2$	Combines Lasso sparsity with Ridge stability.	Yes

- The intercept θ_0 is normally not regularized.
- λ is a hyperparameter selected using validation or cross-validation.
- Larger λ means stronger shrinkage; too much regularization can cause underfitting.

3. Regression Evaluation Metrics

Coefficient of determination (R^2)

$$R^2 = 1 - SSres / SStot$$

$$SSres = \Sigma(y_i - \hat{y}_i)^2 \quad \text{and} \quad SStot = \Sigma(y_i - \bar{y})^2$$

- R^2 measures how much variation in y is explained by the model relative to predicting the mean $\bar{y}$.
- $R^2 = 1$ indicates a perfect fit; $R^2 = 0$ means the model is no better than the mean baseline.
- R^2 can be negative on unseen data when the model performs worse than the mean baseline.

Adjusted R^2

$$\text{Adjusted } R^2 = 1 - (1 - R^2)(m - 1) / (m - n - 1)$$

- m is the number of observations and n is the number of predictors.
- Unlike ordinary R^2, adjusted R^2 penalizes unnecessary predictors.

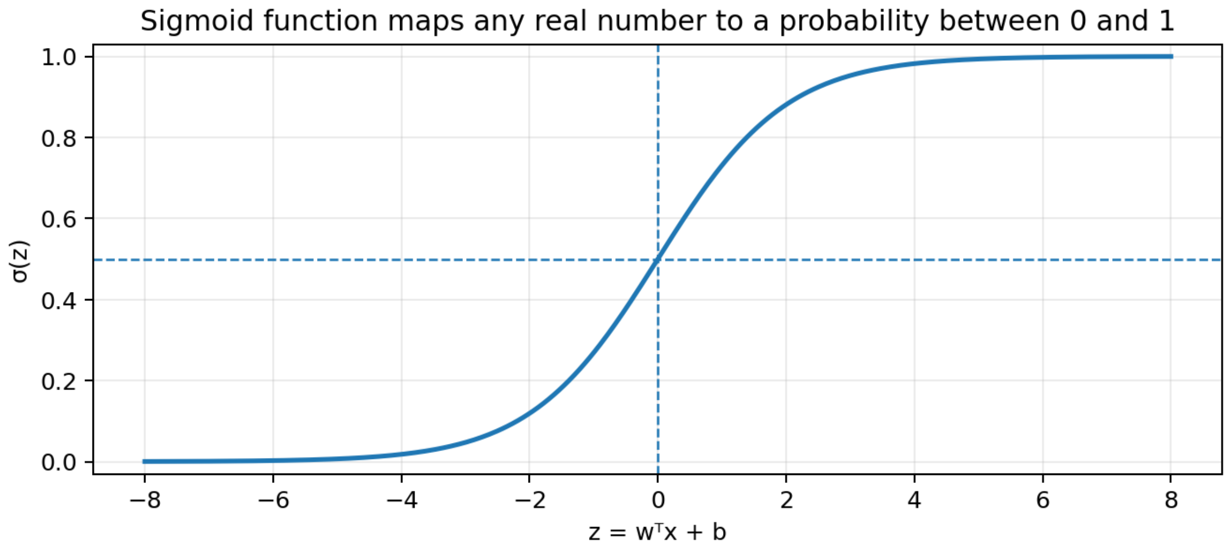
> **Important —** Use R^2 or adjusted R^2 as evaluation metrics, not as the training loss for ordinary linear regression. MSE is commonly minimized during training.

4. Logistic Regression

Logistic regression is primarily used for classification. It predicts the probability that an observation belongs to a class.

Sigmoid function

$$z = w^{T}x + b \qquad \text{and} \qquad \sigma(z) = 1 / (1 + e^{-z})$$



- The output lies between 0 and 1 and can be interpreted as $P(y = 1 \mid x)$.
- A common decision rule is: predict class 1 when probability ≥ 0.5; otherwise predict class 0.
- The threshold can be changed depending on the relative cost of false positives and false negatives.

Binary cross-entropy / log loss

$$J(w,b) = -(1/m) \sum_{i=1}^{m} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

- This loss strongly penalizes confident but incorrect predictions.
- Cross-entropy produces a suitable convex objective for standard binary logistic regression.
- MSE is generally not preferred for logistic regression because it gives less suitable optimization behaviour.

5. Classification Evaluation

Confusion matrix

Actual / Predicted	Predicted positive	Predicted negative
Actual positive	True Positive (TP)	False Negative (FN)
Actual negative	False Positive (FP)	True Negative (TN)

Metric	Formula	Interpretation
Accuracy	(TP + TN) / Total	Overall proportion of correct predictions.
Precision	TP / (TP + FP)	Of predicted positives, how many were correct?
Recall / Sensitivity	TP / (TP + FN)	Of actual positives, how many were found?
Specificity	TN / (TN + FP)	Of actual negatives, how many were correctly rejected?
F1-score	2 × Precision × Recall / (Precision + Recall)	Balance between precision and recall.

Class imbalance — Accuracy alone can be misleading when one class is much more common. Review precision, recall, F1-score and the confusion matrix.

6. Validation and Hyperparameter Tuning

- Training set: used to estimate model parameters.
- Validation set: used to compare models and choose hyperparameters such as λ, tree depth or SVM settings.
- Test set: used once at the end for an unbiased estimate of final performance.

K-fold cross-validation

- Split the available training data into k approximately equal folds.
- Train on k − 1 folds and validate on the remaining fold.
- Repeat until every fold has served once as the validation fold.
- Average the k validation scores to compare models or hyperparameters.

Common choice — k = 5 or k = 10 is common. The final test set should remain untouched during tuning.

7. Support Vector Machine (SVM)

An SVM finds a decision boundary that separates classes while maximizing the margin between them.

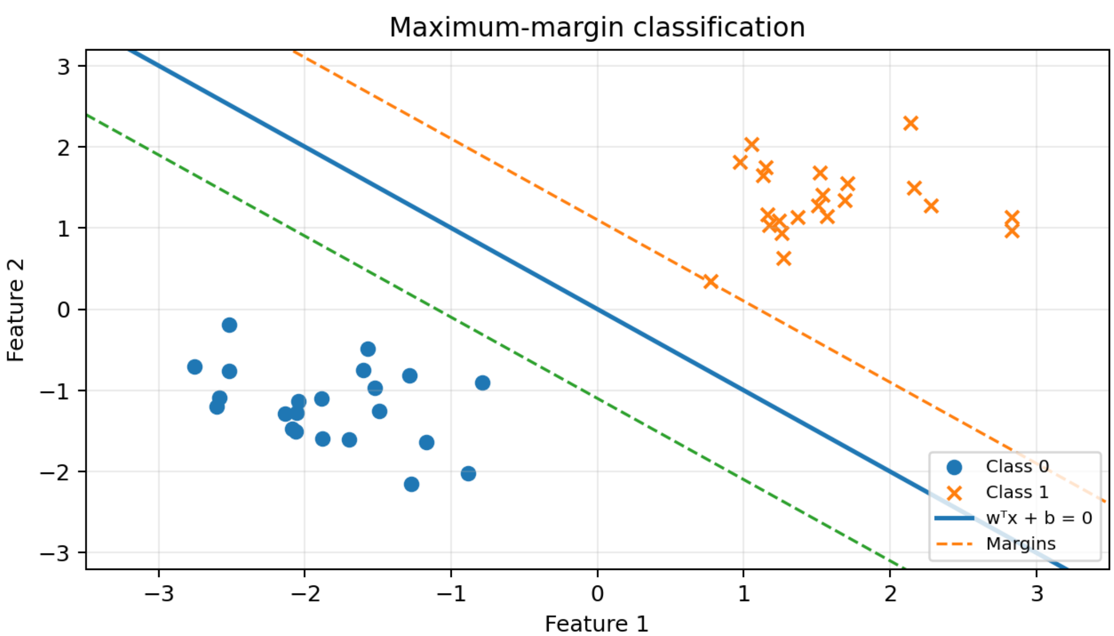
$$\textbf{Decision boundary: } w^T x + b = 0$$

$$\textbf{Margin boundaries: } w^T x + b = +1 \quad \text{and} \quad w^T x + b = -1$$

$$\textbf{Total margin width} = 2 / \|w\|$$

- Support vectors are the training observations closest to the decision boundary.
- These points determine the position and orientation of the separating hyperplane.
- The sign of $w^T x + b$ determines on which side of the boundary a point lies.
- A kernel can create nonlinear boundaries by measuring similarity in a transformed feature space.

Interpretation — Increasing the margin generally improves robustness, but the model must balance margin size against classification errors.



8. Bayes' Theorem

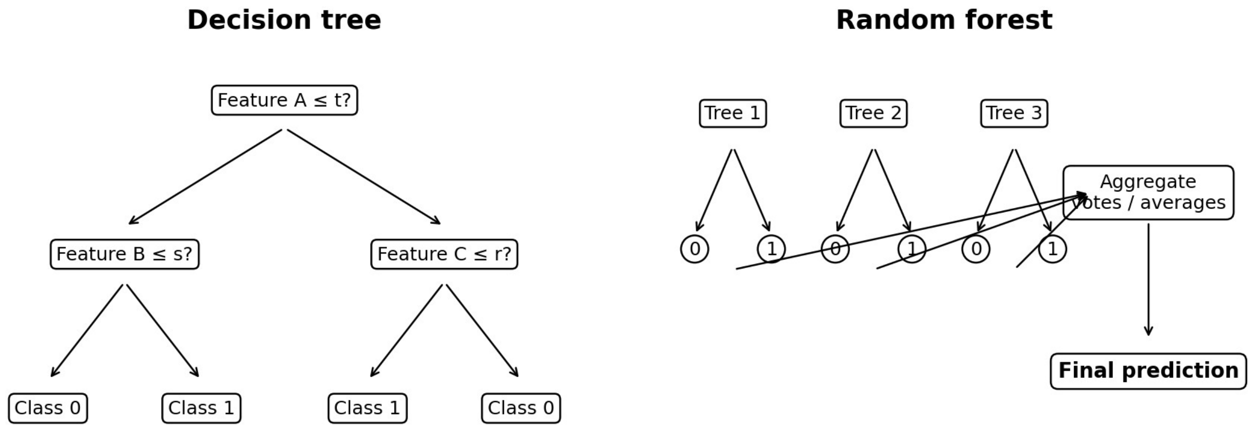
Bayes' theorem updates the probability of a hypothesis after observing new evidence.

$$P(A \mid B) = P(B \mid A) \times P(A) / P(B)$$

Term	Meaning
P(A)	Prior probability of hypothesis A.
P(B | A)	Likelihood of observing evidence B when A is true.
P(B)	Overall probability of the evidence.
P(A | B)	Posterior probability after considering the evidence.

9. Decision Trees

A decision tree repeatedly splits the data into increasingly homogeneous groups using feature-based rules.



Impurity measures

$$\textbf{Entropy:}\quad H(S) = -\sum_k p_k \log_2(p_k)$$

$$\textbf{Gini impurity:}\quad \mathrm{Gini}(S) = 1 - \sum_k p_k^2$$

- A pure node contains observations from only one class and has impurity 0.
- A highly mixed node has higher impurity.
- The best split gives the largest impurity reduction.

Information gain

$$\text{Information Gain} = \text{Impurity(parent)} - \sum_{\text{child}} (n_{\text{child}} / n_{\text{parent}}) \times \text{Impurity(child)}$$

Algorithm	Typical split rule	Key characteristic
ID3	Entropy / information gain	Originally designed for categorical classification features.
CART	Gini impurity for classification; squared error for regression	Uses binary splits and supports classification and regression.

10. Random Forest

A random forest combines many decision trees and aggregates their predictions.

- Each tree is trained on a bootstrap sample: a sample drawn with replacement from the training data.
- At each split, only a random subset of features is considered.
- Classification uses majority voting; regression uses the average prediction.
- Averaging many diverse trees usually reduces variance and overfitting compared with one deep tree.
- Trees can be trained in parallel because they are largely independent.

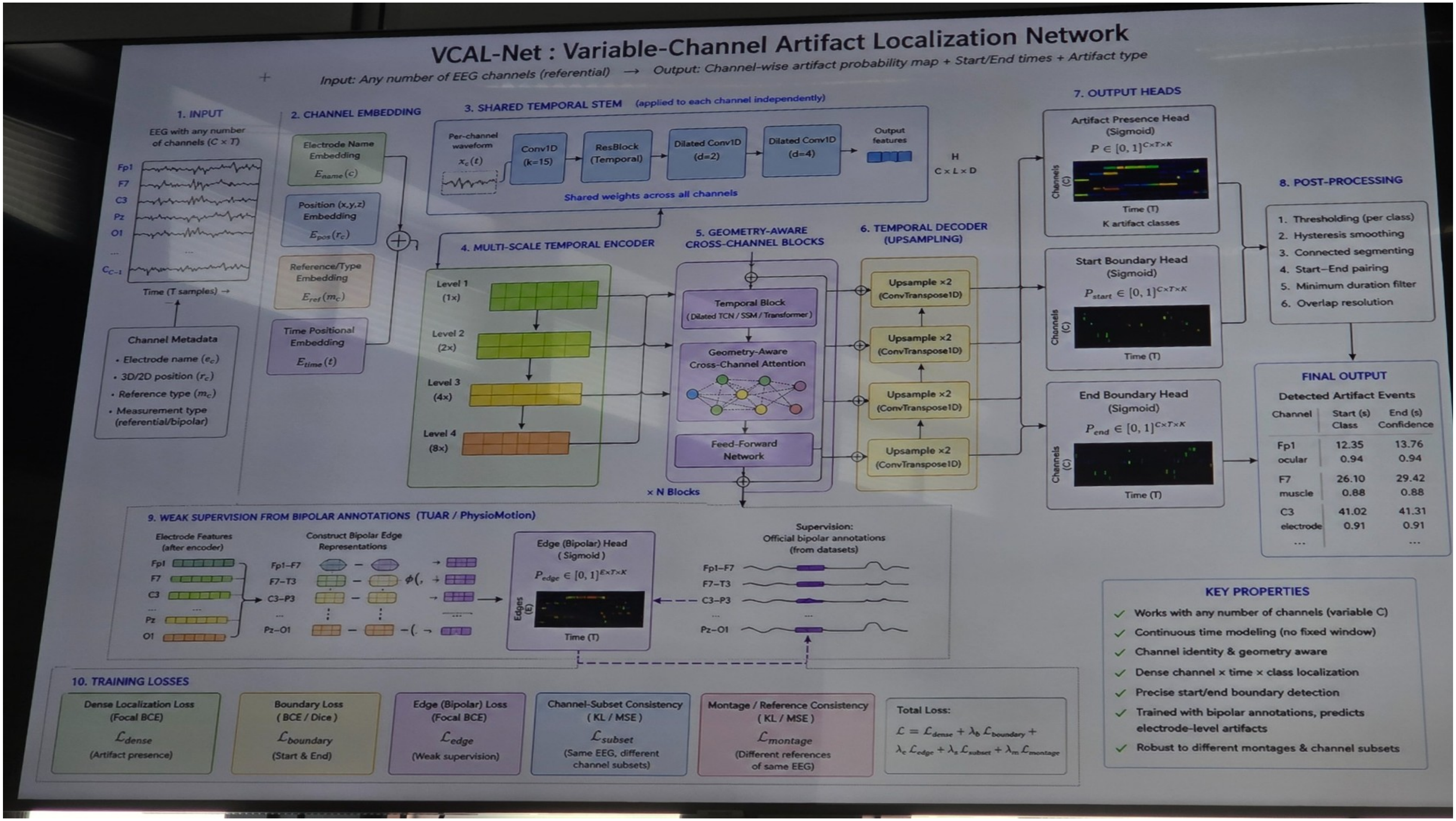
11. Quick Model Comparison

Model	Primary task	Strength	Main limitation
Linear regression	Continuous prediction	Simple, fast and interpretable.	Assumes an approximately linear relationship.
Ridge / Lasso / Elastic Net	Regularized continuous prediction	Controls overfitting; Lasso can select features.	Requires tuning regularization strength.
Logistic regression	Classification	Interpretable probabilities and coefficients.	Linear decision boundary unless features are engineered.
SVM	Classification / regression	Effective in high-dimensional spaces; kernels allow nonlinearity.	Can be harder to tune and scale to very large datasets.
Decision tree	Classification / regression	Easy to explain and handles nonlinear interactions.	A single deep tree can overfit and be unstable.
Random forest	Classification / regression	Strong baseline; lower variance than one tree.	Less interpretable and larger computational footprint.

12. Exam-Ready Summary

Topic	One-line takeaway
Linear regression	Predicts continuous values by minimizing squared residuals.
Gradient descent	Updates parameters in the opposite direction of the loss gradient.
Regularization	Controls model complexity: Ridge shrinks, Lasso selects, Elastic Net combines both.
R^2	Compares residual error with the mean-prediction baseline.
Logistic regression	Uses the sigmoid function and cross-entropy loss for classification.
Cross-validation	Estimates generalization and supports hyperparameter selection.
SVM	Maximizes the margin; support vectors define the boundary.
Decision tree	Chooses splits that reduce entropy or Gini impurity.
Random forest	Averages many randomized trees to reduce variance.

Recommended study sequence — Understand the prediction function first, then the loss, optimization, validation, evaluation metric and finally the method used to control complexity.


VCAL-Net : Variable-Channel Artifact Localization Network
Input: Any number of EEG channels (referential) → Output: Channel-wise artifact probability map + Start/End times + Artifact type
1. INPUT
EEG with any number of channels (C × T)
Fp1
F7
C3
Pz
O1
Time (T samples) →
Channel Metadata
• Electrode name (e_c)
• 3D/2D position (r_c)
• Reference type (m_c)
• Measurement type (referential/bipolar)
2. CHANNEL EMBEDDING
Electrode Name Embedding
$E_{name}(c)$
Position (x,y,z) Embedding
$E_{pos}(r_c)$
Reference/Type Embedding
$E_{ref}(m_c)$
Time Positional Embedding
$E_{time}(t)$
3. SHARED TEMPORAL STEM (applied to each channel independently)
Per-channel waveform
$x_c(t)$
Conv1D (k=15)
ResBlock (Temporal)
Dilated Conv1D (d=2)
Dilated Conv1D (d=4)
Output features
Shared weights across all channels
H
C × L × D
4. MULTI-SCALE TEMPORAL ENCODER
Level 1 (1×)
Level 2 (2×)
Level 3 (4×)
Level 4 (8×)
5. GEOMETRY-AWARE CROSS-CHANNEL BLOCKS
Temporal Block (Dilated TCN / SSM / Transformer)
Geometry-Aware Cross-Channel Attention
Feed-Forward Network
× N Blocks
6. TEMPORAL DECODER (UPSAMPLING)
Upsample ×2 (ConvTranspose1D)
Upsample ×2 (ConvTranspose1D)
Upsample ×2 (ConvTranspose1D)
Upsample ×2 (ConvTranspose1D)
7. OUTPUT HEADS
Artifact Presence Head (Sigmoid)
$P \in [0,1]^{C\times T\times K}$
Channels (C)
Time (T)
K artifact classes
Start Boundary Head (Sigmoid)
$P_{start} \in [0,1]^{C\times T\times K}$
Time (T)
End Boundary Head (Sigmoid)
$P_{end} \in [0,1]^{C\times T\times K}$
Time (T)
8. POST-PROCESSING
1. Thresholding (per class)
2. Hysteresis smoothing
3. Connected segmenting
4. Start–End pairing
5. Minimum duration filter
6. Overlap resolution
FINAL OUTPUT
Detected Artifact Events
Channel | Start (s) Class | End (s) Confidence
Fp1 ocular | 12.35 0.94 | 13.76 0.94
F7 muscle | 26.10 0.88 | 29.42 0.88
C3 electrode | 41.02 0.91 | 41.31 0.91
9. WEAK SUPERVISION FROM BIPOLAR ANNOTATIONS (TUAR / PhysioMotion)
Electrode Features (after encoder)
Construct Bipolar Edge Representations
Fp1–F7
F7–T3
C3–P3
Pz–O1
Edge (Bipolar) Head (Sigmoid)
$P_{edge} \in [0,1]^{E\times T\times K}$
Edges (E)
Time (T)
Supervision: Official bipolar annotations (from datasets)
10. TRAINING LOSSES
Dense Localization Loss (Focal BCE)
$\mathcal{L}_{dense}$
(Artifact presence)
Boundary Loss (BCE / Dice)
$\mathcal{L}_{boundary}$
(Start & End)
Edge (Bipolar) Loss (Focal BCE)
$\mathcal{L}_{edge}$
(Weak supervision)
Channel-Subset Consistency (KL / MSE)
$\mathcal{L}_{subset}$
(Same EEG, different channel subsets)
Montage / Reference Consistency (KL / MSE)
$\mathcal{L}_{montage}$
(Different references of same EEG)
Total Loss:
$\mathcal{L} = \mathcal{L}_{dense} + \lambda_b \mathcal{L}_{boundary} + \lambda_e \mathcal{L}_{edge} + \lambda_s \mathcal{L}_{subset} + \lambda_m \mathcal{L}_{montage}$
KEY PROPERTIES
✓ Works with any number of channels (variable C)
✓ Continuous time modeling (no fixed window)
✓ Channel identity & geometry aware
✓ Dense channel × time × class localization
✓ Precise start/end boundary detection
✓ Trained with bipolar annotations, predicts electrode-level artifacts
✓ Robust to different montages & channel subsets